

DevOps Lab Report

2025

Lab Experiment – 5 & 6

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Program: B.Tech. CSE [Full Stack-AI]

Semester: 6

Batch: Batch2 (Non-Hons)



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Exp – (i) Practice Sheet - Advanced Merging Concepts

1) Fast Forward Merge

The screenshot shows the Visual Studio Code interface with the following components:

- Explorer Panel (Left):** Displays the file structure of the 'devops(upes)' workspace. It includes folders 'lab1', 'lab2', 'lab3', and 'lab4'. Under 'lab4', there is a folder 'advancedMerge' containing 'fastForward.txt' and a file 'README.md'.
- Source Control Panel (Left):** Shows the 'fastForward.txt' file with a commit history. The first commit is '1 fast forward merge feature' by 'You' 25 seconds ago.
- Terminal Panel (Bottom):** Displays the output of a PowerShell session. The commands and their outputs are:
 - `git checkout fast-forward-feature`: Switched to branch 'fast-forward-feature'.
 - `git add .`: Staged changes.
 - `git commit -m "add fast-forward file"`: Committed changes. The commit message is "[fast-forward-feature 4dc0ad7] add fast-forward file". The output shows "1 file changed, 1 insertion(+)" and "create mode 100644 lab4/advancedMerge/fastForward.txt".
 - `git checkout main`: Switched to branch 'main'. The output shows "Your branch is up to date with 'origin/main'".
 - `git merge fast-forward-feature`: Merged the feature branch into main. The output shows "Updating 8f4ab1f..4dc0ad7", "Fast-forward", "lab4/advancedMerge/fastForward.txt | 1 +", "1 file changed, 1 insertion(+)", and "create mode 100644 lab4/advancedMerge/fastForward.txt".
 - `git pull`: Already up to date.
 - `git push`: (Not explicitly shown, but implied by the commit history).
- Activity Bar (Left):** Contains icons for Explorer, Source Control, Run and Debug, Extensions, and Settings.
- Bottom Status Bar:** Shows the current branch 'main', commit hash '0111', and other status information.

2) Non-Fast Forward Merge

The screenshot displays the Visual Studio Code editor interface. On the left, the Explorer sidebar shows a file tree with folders 'lab1', 'lab2', 'lab3', and 'lab4'. Below them is the 'advancedMerge' folder, which contains two files: 'fastForward.txt' and 'nonFastForward.txt'. A 'README.md' file is also visible at the root level.

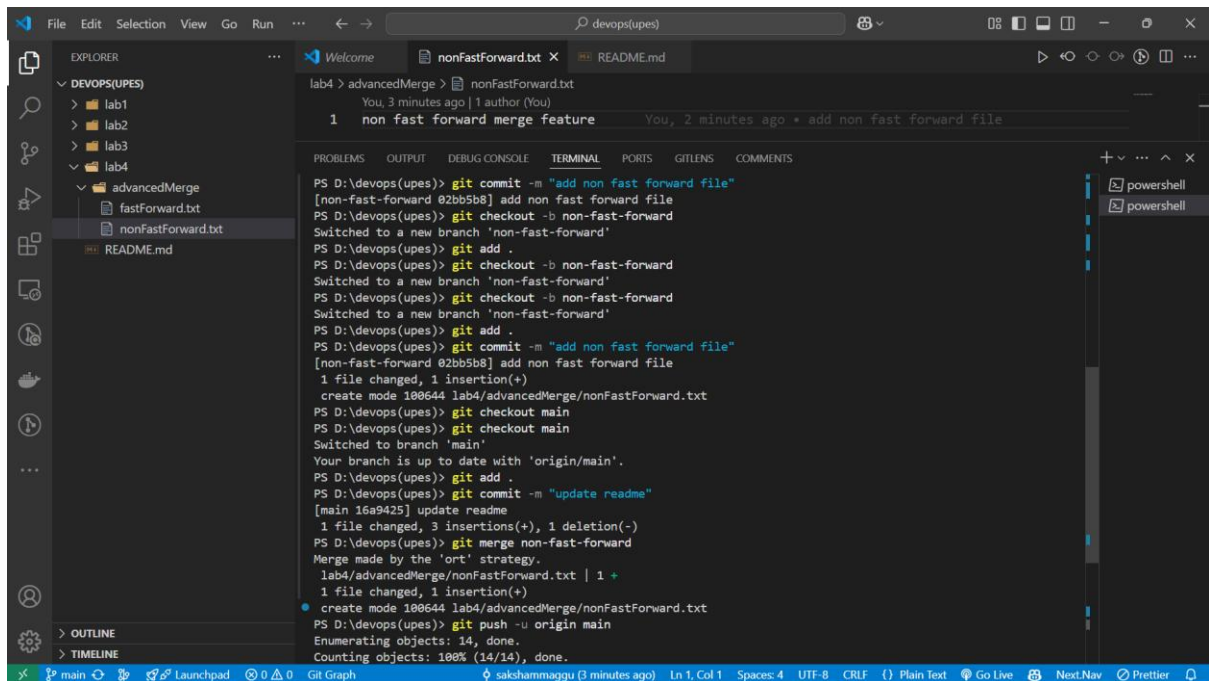
The main editor area has tabs for 'Welcome', 'nonFastForward.txt', and 'README.md'. The 'nonFastForward.txt' tab is active, showing a commit history snippet:

```
lab4 > advancedMerge > nonFastForward.txt
You, 3 minutes ago | 1 author (You)
1 non fast forward merge feature You, 2 minutes ago + add non fast forward file
```

Below the file view, the TERMINAL panel is open, displaying a series of Git commands and their outputs in a PowerShell environment:

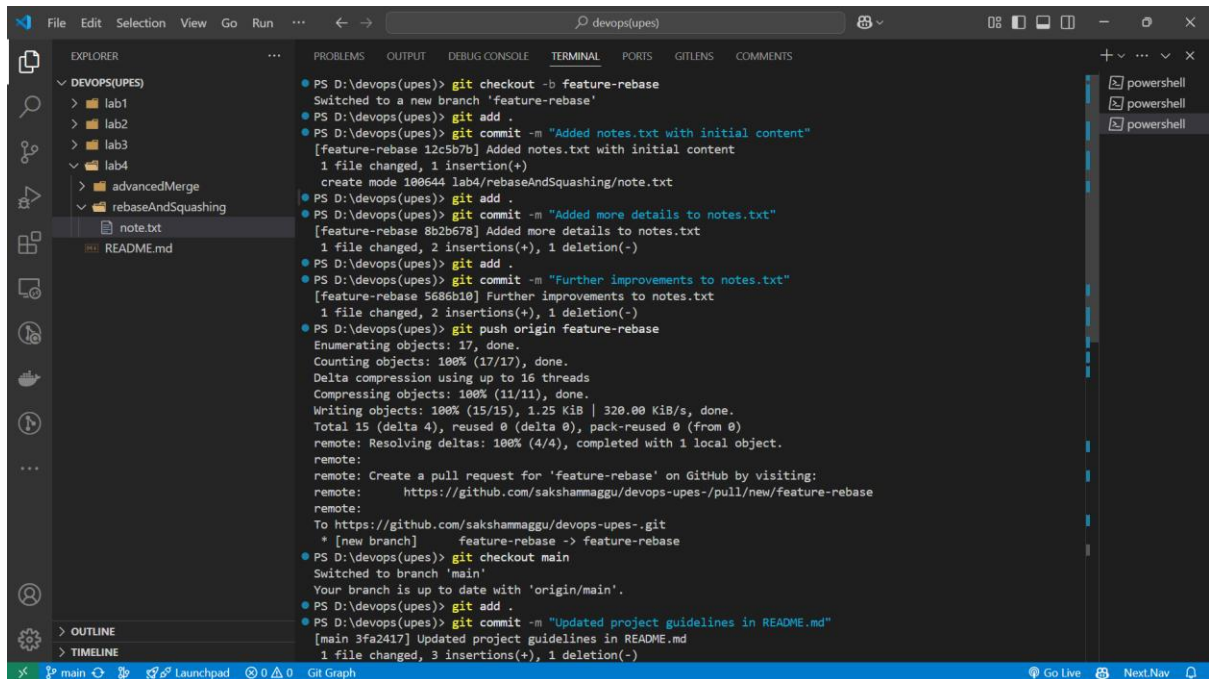
```
PS D:\devops(upes)> git checkout -b non-fast-forward
Switched to a new branch 'non-fast-forward'
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "add non fast forward file"
[non-fast-forward 02bb5b8] add non fast forward file
1 file changed, 1 insertion(+)
create mode 100644 lab4/advancedMerge/nonFastForward.txt
PS D:\devops(upes)> git checkout main
PS D:\devops(upes)> git checkout -b non-fast-forward
Switched to a new branch 'non-fast-forward'
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "add non fast forward file"
[non-fast-forward 02bb5b8] add non fast forward file
1 file changed, 1 insertion(+)
create mode 100644 lab4/advancedMerge/nonFastForward.txt
PS D:\devops(upes)> git checkout -b non-fast-forward
Switched to a new branch 'non-fast-forward'
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "add non fast forward file"
[non-fast-forward 02bb5b8] add non fast forward file
1 file changed, 1 insertion(+)
PS D:\devops(upes)> git checkout -b non-fast-forward
Switched to a new branch 'non-fast-forward'
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "add non fast forward file"
[non-fast-forward 02bb5b8] add non fast forward file
1 file changed, 1 insertion(+)
PS D:\devops(upes)> git checkout -b non-fast-forward
Switched to a new branch 'non-fast-forward'
PS D:\devops(upes)> git add .
```

The status bar at the bottom indicates the current state: 'Git Graph', 'sakshammaagou (3 minutes ago)', 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', 'Plain Text', 'Go Live', 'Next Nav', and 'Prettier'.



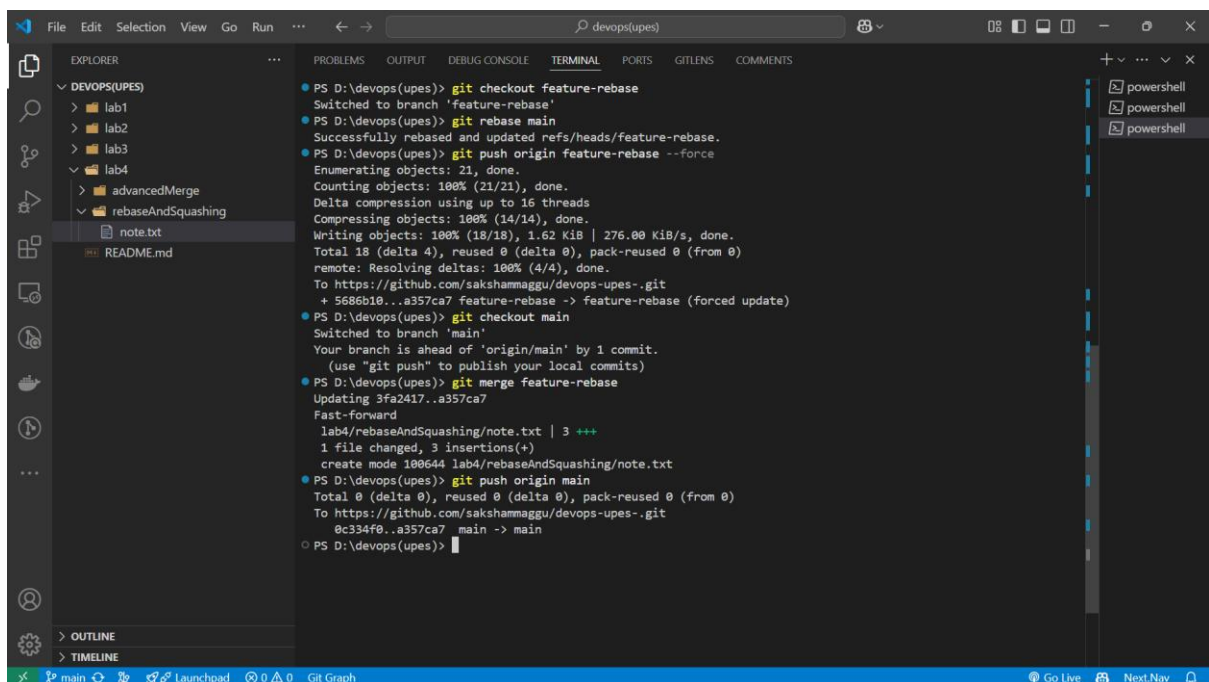
```
PS D:\devops(upes)> git merge non-fast-forward
Merge made by the 'ort' strategy.
 lab4/advancedMerge/nonFastForward.txt | 1 +
 1 file changed, 1 insertion(+)
● create mode 100644 lab4/advancedMerge/nonFastForward.txt
PS D:\devops(upes)> git push -u origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 973 bytes | 973.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/sakshammaggu/devops-upes-.git
 4dc0ad7..0c334f0  main -> main
● branch 'main' set up to track 'origin/main'.
PS D:\devops(upes)> git pull
○ Already up to date.
PS D:\devops(upes)>
```

Exp – (ii) Practice Sheet – Rebasing and Commit squashing



The screenshot shows the VS Code interface with the Explorer, Problems, Output, Debug Console, and Terminal tabs. The Explorer shows a project structure with folders lab1, lab2, lab3, lab4, advancedMerge, and rebaseAndSquashing. The rebaseAndSquashing folder contains note.txt and README.md. The Terminal shows the following commands and output:

```
PS D:\devops(upes)> git checkout -b feature-rebase
Switched to a new branch 'feature-rebase'
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "Added notes.txt with initial content"
[feature-rebase 12c5b7b] Added notes.txt with initial content
1 file changed, 1 insertion(+)
create mode 100644 lab4/rebaseAndSquashing/note.txt
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "Added more details to notes.txt"
[feature-rebase 8b2b678] Added more details to notes.txt
1 file changed, 2 insertions(+), 1 deletion(-)
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "Further improvements to notes.txt"
[feature-rebase 5686b10] Further improvements to notes.txt
1 file changed, 2 insertions(+), 1 deletion(-)
PS D:\devops(upes)> git push origin feature-rebase
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 16 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 1.25 KiB | 320.00 KiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-rebase' on GitHub by visiting:
remote:   https://github.com/sakshammaggu/devops-upes-/pull/new/feature-rebase
remote:
To https://github.com/sakshammaggu/devops-upes-.git
 * [new branch]      feature-rebase -> feature-rebase
PS D:\devops(upes)> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\devops(upes)> git add .
PS D:\devops(upes)> git commit -m "Updated project guidelines in README.md"
[main 3fa2417] Updated project guidelines in README.md
1 file changed, 3 insertions(+), 1 deletion(-)
```



The screenshot shows the VS Code interface with the Explorer, Problems, Output, Debug Console, and Terminal tabs. The Explorer shows the same project structure as the previous screenshot. The Terminal shows the following commands and output:

```
PS D:\devops(upes)> git checkout feature-rebase
Switched to branch 'feature-rebase'
PS D:\devops(upes)> git rebase main
Successfully rebased and updated refs/heads/feature-rebase.
PS D:\devops(upes)> git push origin feature-rebase --force
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 16 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (18/18), 1.62 KiB | 276.00 KiB/s, done.
Total 18 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/sakshammaggu/devops-upes-.git
+ 5686b10...a357ca7 feature-rebase -> feature-rebase (forced update)
PS D:\devops(upes)> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
PS D:\devops(upes)> git merge feature-rebase
Updating 3fa2417..a357ca7
Fast-forward
 lab4/rebaseAndSquashing/note.txt | 3 +++
1 file changed, 3 insertions(+)
create mode 100644 lab4/rebaseAndSquashing/note.txt
PS D:\devops(upes)> git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sakshammaggu/devops-upes-.git
0c334f0..a357ca7 main -> main
PS D:\devops(upes)>
```